Department of Instrumentation & Control Engineering • Minor Project Phase-II • 2023–2027

# Smart Pesticide Control System

## Driven by Plant Infection Analysis

2026

*AI-IoT Based Severity-Aware Pesticide Spraying for Sustainable Agriculture*

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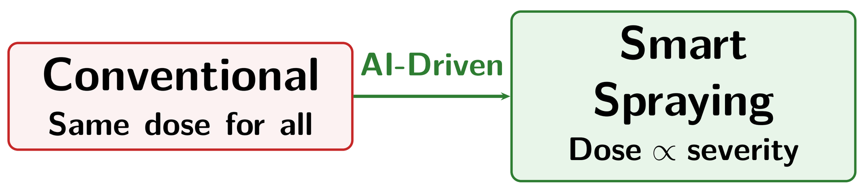
## 1 Motivation & Problem Statement

Agriculture still relies on **uniform pesticide spraying**—chemicals applied across the field without considering individual plant health, leading to **chemical wastage**, increased cost, soil contamination, and unnecessary exposure of healthy plants.

> *"How can we spray pesticide only where needed and in proportion to disease severity?"*

**Core challenge:** Build a closed-loop system connecting image-based severity detection with proportional pesticide actuation.




## 2 Research Gap

Most existing systems handle detection and spraying *separately*:

- ML systems detect disease but **do not control spray quantity**
- Robotic sprayers are costly and not severity-aware
- Binary infected/not-infected decisions are insufficient
- No affordable system links **detection with proportional actuation**

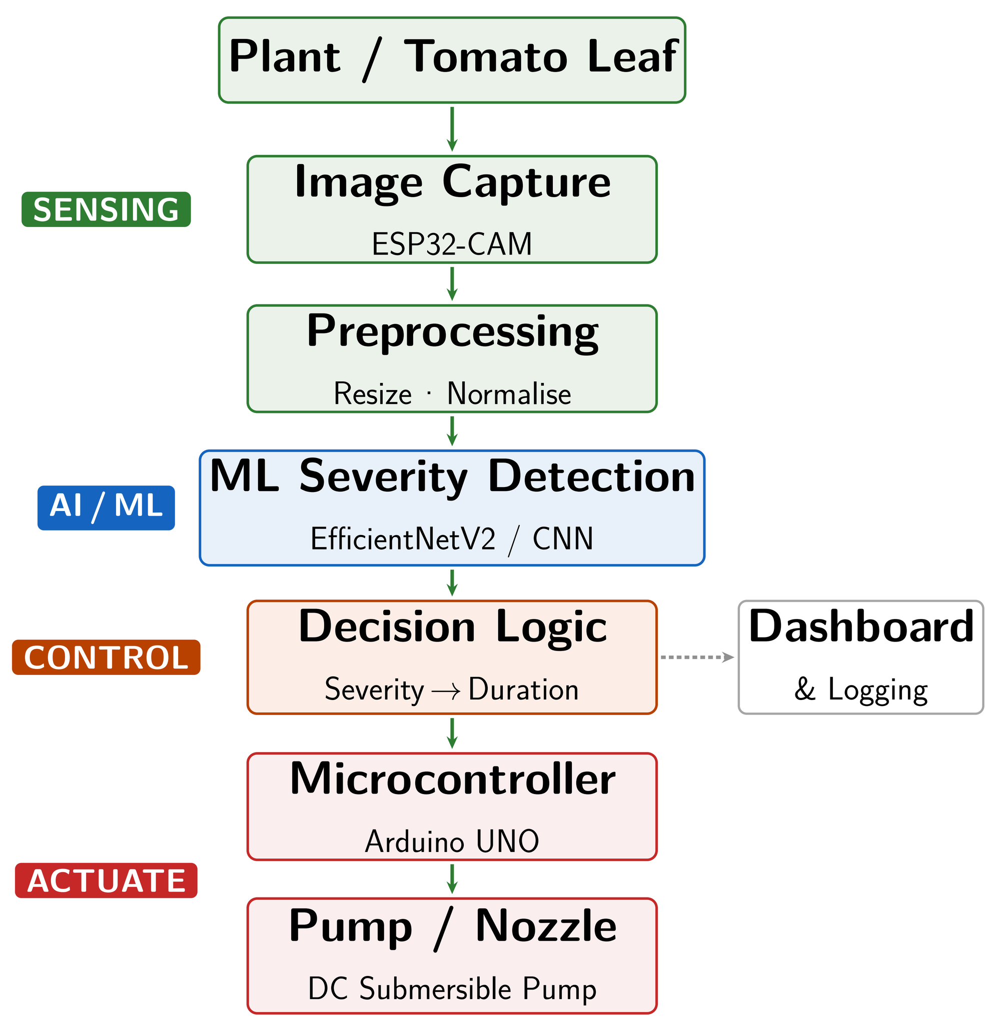
| Existing Direction | Limitation |
|---|---|
| ML disease detection | Stops at classification |
| Robotic sprayers | Expensive, not severity-aware |
| AI-IoT farming | Limited proportional control |
| **Our system** | **Detection + proportional actuation** |

## 3 Objectives

1. Capture tomato leaf images using ESP32-CAM module
2. Classify infection severity: *Healthy / Moderate / Severe*
3. Map severity to calibrated pump actuation duration
4. Implement microcontroller-based pump control
5. Display infection status via RGB LED and 7-segment display
6. Reduce pesticide usage vs. conventional uniform spraying
7. Build a low-cost, working prototype for academic use

## 4 System Architecture

The system forms a **closed-loop pipeline** from plant image acquisition to physical spray actuation:




## 5 Decision Logic

The decision layer converts predicted infection severity into pump ON time, making the system **severity-aware** rather than spraying all plants equally.

| Infection | Class | Action | Duration |
|---|---|---|---|
| Low (<30%) | Healthy | No spray | **0 s** |
| Medium (30–60%) | Moderate | Short spray | **3 s** |
| High (>60%) | Severe | Extended spray | **7 s** |

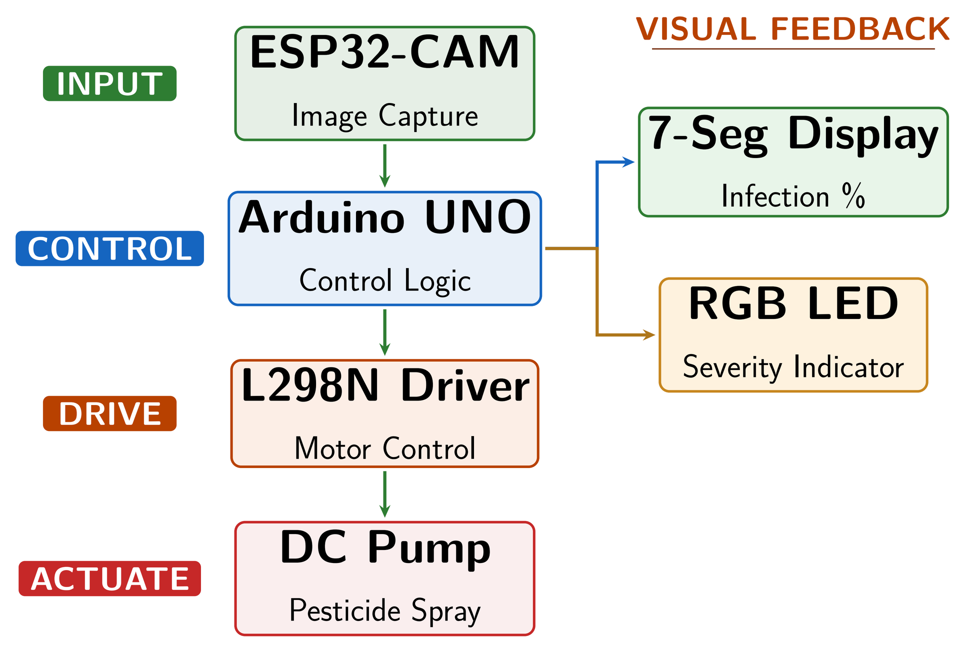
**Healthy** 0 sec · **Moderate** 3 sec · **Severe** 7 sec

## 6 Prototype Implementation

A working prototype demonstrates the complete closed-loop pipeline from image capture to pump actuation.

| Component | Role |
|---|---|
| ESP32-CAM | Captures leaf image |
| Arduino UNO | Control logic & hardware interface |
| RGB LED | Infection severity indicator |
| 7-Segment Display | Shows infection percentage |
| L298N Motor Driver | Drives the DC pump |
| DC Submersible Pump | Pesticide spraying actuation |

**Hardware Signal Flow:**




## 7 Expected Performance & Pesticide Saving

| Metric | Target |
|---|---|
| Classification accuracy | $\geq$ 90% |
| End-to-end latency | $\leq$ 5 s |
| Pesticide reduction | $\geq$ 40% |
| Theoretical saving (mixed case) | $\approx$ 57% |

**Pesticide Reduction Analysis** (sample 10-plant mixed-health test):

| Type | Count | Uniform | Smart | Saving |
|---|---|---|---|---|
| Healthy | 4 | 28 s | 0 s | 28 s |
| Moderate | 3 | 21 s | 9 s | 12 s |
| Severe | 3 | 21 s | 21 s | 0 s |
| **Total** | **10** | **70 s** | **30 s** | **40 s** |

Smart spraying reduces total spray time from **70 s** to **30 s**—a **≈57% reduction** in pesticide use.

## 8 Applications & Future Scope

### Applications

- Small & marginal farms
- Greenhouses & polyhouses
- Agricultural research stations
- Precision agriculture

### Future Scope

- Multi-crop support
- Autonomous robot cart
- Cloud dashboard & mobile app
- Edge AI inference

## 9 Conclusion

- Integrates **computer vision, ML, and embedded control** for severity-aware pesticide spraying
- Healthy plants receive **zero spray**; dose scales proportionally with infection severity
- Prototype validates the closed-loop pipeline: **detection** → **decision** → **actuation**
- Achieves theoretical **≈57% pesticide reduction** vs. uniform spraying

**Spray only where needed, in proportion to severity.**
AI + IoT for sustainable precision agriculture.

## Keywords & References

**Keywords:** Precision Agriculture · AI · IoT · Computer Vision · ESP32-CAM · Arduino · EfficientNetV2 · Smart Spraying · Tomato Leaf Disease

**References:** [1] PlantVillage dataset, Kaggle. [2] Tan & Le, EfficientNetV2, ICML 2021. [3] IoT precision agriculture, MDPI Sensors. [4] CNN leaf disease detection, IEEE Access.